

Two points, one parity

APN fibres as $\mathbb{Z}/2$ -torsors, and why H^1 vanishes, read slowly

Abstract

One arithmetic fact, dressed in type theory.

- An additive homomorphism from a finite group of *odd order* to $\mathbb{Z}/2$ is the zero map.

That is the whole engine. We watch it run a loop that is genuinely type/topos-theoretic at every step except the last:

$$2\text{-element fibre} \rightarrow \mathbb{Z}/2\text{-torsor} \rightarrow 1\text{-cocycle} \rightarrow \text{coboundary},$$

and the only reason the last arrow exists is that $|F^\times| = 2^n - 1$ is *odd*.

The bookkeeping (torsor, cocycle, coboundary, H^1) is category- and type-theoretic; the punchline is a one-line parity count. We read each step as *math* $\leftrightarrow$ *Lean* $\leftrightarrow$ *a typing judgement*.

The load-bearing lemmas below compile in Lean 4 with Mathlib, with no `sorry` and only the standard axioms (`ToposH1Triviality.lean`).

The object: a derivative fibre

Fix a finite field $F = \text{GF}(2^n)$ and a map $f : F \rightarrow F$. The **derivative** in direction a and its **fibre** over b are

$$D_a f(x) = f(x + a) - f(x), \quad \text{fib}(f, a, b) = \{x : D_a f(x) = b\}.$$

```
def D (f : F → F) (a x : F) : F := f (x + a) - f x
def fib (f : F → F) (a b : F) : Finset F :=
  Finset.univ.filter (fun x => D f a x = b)
def N (f : F → F) (a b : F) : ℕ := (fib f a b).card
```

APN (almost perfect nonlinear) means every non-zero direction has fibres of size at most 2:

```
def IsAPN (f : F → F) : Prop := ∀
  a : F, a ≠ 0 → ∀ b : F, N f a b ≤ 2
```

In Lean’s type theory the predicate $x \mapsto (D_a f(x) = b)$ is a map $F \rightarrow \Omega$ with $\Omega = \text{Prop}$, $|\Omega| = 2$. The fibre is the subobject it classifies. “APN” is exactly “no subobject is larger than Ω ”. We pick up that thread in the companion note *APN Fibres and the Subobject Classifier*; here we go one floor higher and ask what the group $\mathbb{Z}/2$ is doing inside each two-point fibre.

1 Each non-empty fibre is a $\mathbb{Z}/2$ -torsor

In characteristic 2 the shift $x \mapsto x + a$ is a fixed-point-free involution, and it maps every fibre to itself:

```
theorem fiber_swap [CharP F 2] (f : F → F) (hf : IsAPN f)
  (a : F) (ha : a ≠ 0) (b : F) (x : F) (hx : x ∈ fib f a b) :
  x + a ∈ fib f a b

theorem fiber_swap_involution [CharP F 2] (a x : F) :
  x + a + a = x
```

So $\mathbb{Z}/2$ acts on each fibre: the non-trivial element acts by $x \mapsto x + a$, and $(x + a) + a = x$ confirms it is an involution. The action is free (no fixed point, since $a \neq 0$) and, on a *non-empty* APN fibre, it is transitive: a fibre has at most 2 elements, and the two are interchanged.

A free, transitive action of a group G on a non-empty set is exactly a **G -torsor**: a copy of G that has forgotten where its identity is.

$$\begin{array}{ccc} x & \xrightleftharpoons{+a} & x + a \\ & \cong & \\ 0 & \xrightleftharpoons{+1} & 1 \end{array}$$

The automorphism group of such a fibre is therefore $\mathbb{Z}/2$ exactly. In Lean: any fibre-preserving injection moves each point to itself or to its partner, nothing else.

```
theorem fiber_aut_le_two [CharP F 2] (f : F → F) (hf : IsAPN f)
  (a : F) (ha : a ≠ 0) (b : F) (hb : N f a b = 2) φ
  ( : F → F) (φh : ∀ x ∈ fib f a b, φ x ∈ fib f a b)
  (φh_inj : Function.Injective φ) : ∀
  x ∈ fib f a b, φ x = x ∨ φ x = x + a
```

$\text{Aut}(\text{fibre}) = \mathbb{Z}/2$. A torsor is not canonically a group — there is no preferred “0” among $\{x, x + a\}$ — but its symmetries form the group $\mathbb{Z}/2$ on the nose. That distinction (torsor vs. group) is precisely what makes the next section non-trivial: a *coherent* choice of basepoint across all directions a is a section, and sections can be obstructed.

2 Comparing two APN functions: a 1-cocycle

Suppose f and g are APN and share their branch structure: for each direction a there is a relabelling σ_a of the codomain carrying f ’s fibres onto g ’s,

$$D_a g(x) = \sigma_a(D_a f(x)).$$

Because each fibre is a $\mathbb{Z}/2$ -torsor, the *ambiguity* in σ_a — whether it preserves or swaps a chosen basepoint — is a single bit, recorded by a parity function:

```
def fiberParity σ ( : F → F ) ( b : F ) : ZMod 2 :=
  if σ b = b then 0 else 1
```

Composing relabellings along $a + b = c$ multiplies torsors, and the parities *add*. Packaged over all directions, the family $\{\sigma_a\}$ satisfies the 1-cocycle identity — this is the structural heart of a descent datum:

```
structure DescentDatum (f g : F → F) where σ
  : F → F → F
  bij : ∀ a : F, a ≠ 0 → Function.Bijective σ ( a )
  compat : ∀ a : F, a ≠ 0 → ∀ x : F, D g a x = σ a ( D f a x )
  cocycle : ∀ a b : F, a ≠ 0 → b ≠ 0 → a + b ≠ 0 → ∀
    x : F, σ (a + b) ( D f (a + b) x ) = σ
      a ( D f a (x + b) ) + σ b ( D f b x )
```

Read σ as a function $F^\times \rightarrow \text{Aut}(\text{fibre}) = \mathbb{Z}/2$ once we project to parities. The `cocycle` field is the familiar 1-cocycle law $c(ab) = c(a) + c(b)$ (additive, char 2); a *constant* relabelling is a 1-coboundary. “ f and g differ by an honest, direction-independent relabelling” = “the cocycle is a coboundary” = “the class in H^1 is zero”.

$$\begin{array}{ccc} F^\times & \xrightarrow{c=\text{parity} \circ \sigma} & \mathbb{Z}/2 \\ & \searrow 0 & \parallel \\ & & \mathbb{Z}/2 \end{array} \quad c(a+b) = c(a) + c(b).$$

3 The vanishing: odd order kills the parity

Here is the only arithmetic in the whole story. The group of directions is $F^\times$, of order

$$|F^\times| = 2^n - 1,$$

which is **odd** for every $n \geq 1$. And an additive homomorphism from an odd-order group to $\mathbb{Z}/2$ has no choice but to vanish:

```
theorem hom_trivial_odd_order_add
  (G : Type*) [AddCommGroup G] [Fintype G]
  (hG_odd : Odd (Fintype.card G))
  (c : G →+ ZMod 2) : ∀
    g : G, c g = 0
```

Why it is true (three lines)

For any g , write $m = |G|$. Then

$$m \cdot c(g) = c(m \cdot g) = c(0) = 0,$$

because $m \cdot g = 0$ in a group of order m (Lagrange). In $\mathbb{Z}/2$, adding an element to itself an *odd* number of times returns the element: $m \equiv 1 \pmod{2}$, so $m \cdot c(g) = c(g)$. Combining, $c(g) = 0$. ■

Mechanically the Lean proof reads it through orders: $2 \cdot c(g) = 0$ forces $\text{addOrderOf } c(g) \mid 2$, while $\text{addOrderOf } c(g) \mid |G|$ is odd; the only common possibility is order 1, i.e. $c(g) = 0$.

$$\frac{2 \cdot c(g) = 0 \Rightarrow \text{ord } c(g) \mid 2 \quad \text{ord } c(g) \mid |G| \quad |G| = 2^n - 1 \text{ odd}}{\text{ord } c(g) = 1 \Rightarrow c(g) = 0} \text{ PARITY}$$

The multiplicative phrasing is the same fact wearing 1 for its identity:

```
theorem hom_trivial_odd_order_mul
  (G : Type*) [CommGroup G] [Fintype G]
  (hG_odd : Odd (Fintype.card G))
  (c : G →* ZMod 2) : ∀
    g : G, c g = 1
```

and, named for what it means, H^1 vanishes — every 1-cocycle is a coboundary:

```

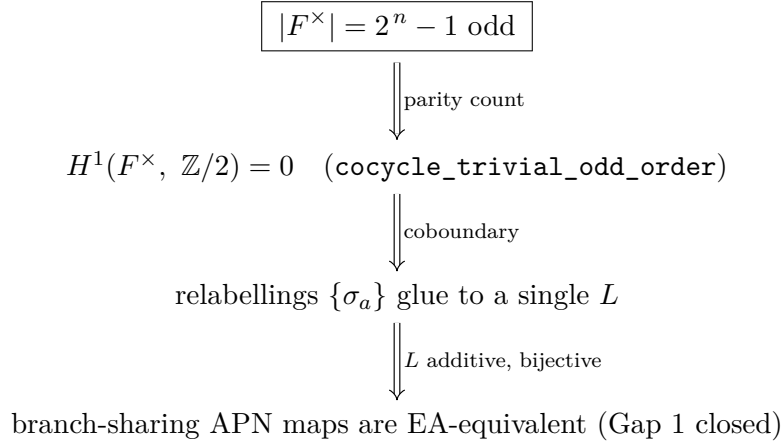
theorem cocycle_trivial_odd_order
  (G : Type*) [AddCommGroup G] [Fintype G]
  (hG_odd : Odd (Fintype.card G))
  (c : G →+ ZMod 2) :∀
    g : G, c g = 0 := hom_trivial_odd_order_add G hG_odd c

```

$H^1(F^\times, \mathbb{Z}/2) = 0$, for *every* n , because $|F^\times| = 2^n - 1$ is always odd. The cohomology is genuinely the right language — torsors, cocycles, coboundaries are not analogies here — yet the vanishing is settled by a single parity count, the same “odd order admits no non-trivial map to $\mathbb{Z}/2$ ” that powers Feit–Thompson-flavoured arguments in miniature.

4 What this does and does *not* settle

The vanishing of H^1 trivialises the **cocycle ambiguity** (“Gap 1”): if two APN functions share a branch, the per-direction relabellings σ_a can be glued into one direction-independent relabelling L .



But notice what the parity argument is *blind* to: it never used the parity of n . The obstruction it removes vanishes for all n . So the genuine even- n difficulty in the APN conjecture cannot live here — it lives in **Gap 2**, the classification of which branches occur at all. The torsor/ H^1 layer is exactly the part of the problem that type theory makes *free*.

One-screen summary

layer	statement	engine
fibre	$\{x, x + a\}$ is a $\mathbb{Z}/2$ -torsor	free involution, char 2
Aut	symmetries of a fibre = $\mathbb{Z}/2$	<code>fiber_aut_le_two</code>
compare	$\{\sigma_a\}$ is a 1-cocycle	<code>DescentDatum.cocycle</code>
vanish	$H^1(F^\times, \mathbb{Z}/2) = 0$	$ F^\times = 2^n - 1$ odd

`fiber_swap/fiber_aut_le_two` $\Rightarrow$ `DescentDatum` $\Rightarrow$ `cocycle_trivial_odd_order`
 $\Rightarrow$ Gap 1 closed.

Reproducibility

The torsor and H^1 -vanishing lemmas (`fiber_swap`, `fiber_swap_involution`, `fiber_aut_le_two`, `fiberParity`, `hom_trivial_odd_order_add`, `hom_trivial_odd_order_mul`, `cocycle_trivial_odd_order`) compile in Lean 4 with Mathlib, with no sorry and only the standard axioms, in `ToposH1Triviality.lean`. The downstream gluing statements (`descent_is_effective`, `omega_descent`) are stated there as the remaining *Gap 2* content and are not used by anything above.